

The Triadic Measurement Substrate

$$\bar{B} = 3\bar{\alpha}$$

Volume 1

Paper 4: Hierarchies, Self-Modeling Measurement Nodes, Multi-Subsystem Dynamics, and Novelty Exhaustion

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Paper 4

Hierarchies, Self-Modeling Measurement Nodes, Multi-Subsystem Dynamics, and Novelty Exhaustion

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Status of Results

This paper distinguishes four categories of claim, each labeled where it appears in the text.

Derived. Results that follow from the five axioms, the Principle of Generative Economy, and the structures established in Papers 1 through 3 by explicit construction or proof. The recursive coarse-graining operator $G^{(k)}$ at arbitrary hierarchical depth as the recursive application of Paper 3's G under axiom scale-independence (§ 2.1); the scale-separation requirement $L_p^{(k)} \gg L_p^{(k-1)}$ and $T_p^{(k)} \gg T_p^{(k-1)}$ as forced by the meta-level coherence threshold (§ 2.2); the existence and finiteness of the hierarchy ceiling $k_{\max}$ for any bounded region (§ 2.3); the Russian-doll floor at L_p for downward nesting (§ 2.4); the structural asymmetry between lossy coarse-graining and generative downward nesting, establishing per-level independence of configuration spaces (§ 2.5.1); the three growth modes (annexation, imprinting, incorporation) with their respective geometric, informational, and structural ceilings (§ III); self-modeling as a structural property of every subsystem at $k \geq 1$, with the three conditions of self-modeling jointly satisfied by the (C_D, C_O, C_C) trichotomy plus the Imprinting Theorem (§ 4.2); the identification of every subsystem at $k \geq 1$ as a measurement node at its level (§ 4.3); higher-order self-modeling at $k \geq 2$ including self-modeling-of-self as a derived consequence of the recursion (§ 4.6); meta-meta-confirmation as the same triadic structure applied at the next level, with the polarization-biased resolution rule inherited unchanged (§ 5.2); the coupling-timescale constraint $T_p^{(k+1)} \geq \sqrt{2} \cdot (L_p(k+1)/L_p(k)) \cdot T_p^{(k)}$ from the requirement of coherent meta-meta-confirmation (§ 5.3); three substrate-level stressors (mortality, predation, starvation) with substrate mechanisms and level-($k+1$) signatures (§ 5.5–§ 5.7); the overdetermination of subsystem formation by Borel-Cantelli inevitability, PGE preference, and flux-balance forcing (§ 5.1); inter-region and intra-region diversity as derived consequences of substrate-level dynamics (§ 5.9); the finiteness of accessible configuration space $\Sigma(R)$ for any bounded region (§ 6.1); monotonic decay of the novelty production rate $v(R, \tau)$ as Σ_{sampled} approaches $\Sigma(R)$ (§ 6.2); the structural exhaustion problem at $\tau > \tau^*(R)$, with Axioms 3 and 4 jointly forcing reinterpretation as the only available continuation (§ 6.6); and the upper bound $c^{(k+1)} \leq (1/\sqrt{2}) \cdot c^{(k)}$ on the hierarchical lightspeed dispersion as a structural law of the recursive FCC face-diagonal-to-edge ratio at every hierarchical level (§ 7.4 sharpened with reference to Paper 5 § 7.2 and Paper 3 § 4.5, holding in all regimes; the tight-coupling case saturates equality) are derived in this sense.

Structural correspondence. The exhaustion threshold $v * (R)$ of § 6.3 and its full symbolic expansion in Appendix A are derivations of the structural form, not of the numerical value. The functional dependences — formation rate set by sampling rate, dissolution rate set by per-level mortality plus predation plus starvation contributions, exhaustion onset when the former drops below the latter — are forced by inherited machinery. The geometric proportionality constants $G_{\text{pred}}^{(k)}$ and $G_{\text{starv}}^{(k)}$, together with the $C(4,2) = 6$ combinatorial factor inherited from Paper 3, are stated as worst-case bounds whose exact refinement is reserved for Paper 5. The threshold's qualitative behavior is forced by the derivation; its quantitative pinning is not. The hierarchical lightspeed dispersion spectrum (§ 7.3) is structurally constrained by the upper bound but unbounded below; the explicit $c^{(k)}$ sequence depends on combinatorial enumeration deferred to Paper 5.

Predictions / Conjectures. Falsifiable claims that follow from the framework but whose closed forms are not derived within this paper. The fully refined $\beta = f(d, K_{\psi,\text{local}}, K_{\psi,\text{surround}}, K_{\psi,\text{subsystem}}, k)$ of § 8.1 specifies the functional dependences across all five arguments; f 's explicit form is deferred. The cross-level relativistic phenomena flagged in § 8.2 (variable apparent propagation, frame-relative timing, embedded-medium dilation) are structural consequences of the $c^{(k)}$ dispersion and the field reinterpretation protocol; their derivation as the substrate-level origins of relativistic phenomena requires the closed-form $c^{(k)}$ values and the reinterpretation protocol of Paper 5. The identification of our measured speed of light as $c^{(k)}$ for some hierarchical level k determined by which level our universe occupies (§ 7.6) is a structural prediction whose empirical content depends on Paper 5's simulation work.

Deferred derivations carried into Paper 5. Several results are stated structurally in this paper and require Paper 5's simulation work for closure: the closed-form $c^{(k)}$ values at successive levels, the exact integer $L_p^{(k)}$ values, the numerical evaluation of $G_{\text{pred}}^{(k)}$ and $G_{\text{starv}}^{(k)}$, the full closed form of f , the cross-level relativistic derivation, the reinterpretation protocol that takes the substrate from exhaustion to fundamental fields, the identification of those fields and their initial conditions, and the Big Bang correspondence. Paper 4 establishes the structural framework within which these closures occur; Paper 5 supplies the closures. The four-paper sequence of Volume 1 is the structural derivation; Paper 5 is the numerical and protocol pinning that converts the structural results into specific empirical content.

This four-way labeling applies throughout the paper. Every claim in the body text falls into exactly one category.

Preface to Paper 4

This paper is the fourth installment in Volume 1 of the Triadic Measurement Substrate program. Volume 1, Paper 1 established the geometric and ontological architecture of the substrate. Volume 1, Paper 2 developed the substrate's computational dynamics. Volume 1, Paper 3 established how computation emerges from informational substrate via surround-driven repair, polarization-coherence bistability, recursive coarse-graining, and the first computing subsystem. Paper 4 takes all three as given and develops the macroscopic consequences promised at Paper 3 § 6.5: the recursive hierarchies of subsystems, the conditions under which a subsystem develops an internal model of its environment, the dynamics of multiple subsystems sharing a meta-interstice environment, the eventual exhaustion of accessible novelty as a substrate region completes its configuration space, and the hierarchical lightspeeds $c^{(k)}$ developed in detail. As in Papers 1 through 3, no new axioms are introduced. No new free numerical parameters are introduced. The Principle of Generative Economy continues to govern every selection.

Paper 4 is the connective tissue of Volume 1. Paper 3 derived the first computing subsystem; Paper 5 will develop the reinterpretation protocol by which a substrate region whose accessible configuration space has exhausted is read under a new protocol, producing the fundamental fields and initial conditions of a

simulated universe. Between these two endpoints lies the work of Paper 4: showing how subsystems multiply, hierarchize, self-model, stress one another, diversify, and ultimately exhaust their local configuration spaces — the structural sequence that makes reinterpretation both necessary and possible.

Abstract. *Building on the first computing subsystem of Paper 3, we develop the macroscopic consequences of subsystem dynamics. Six principal results are derived. First, the recursive coarse-graining operator G of Paper 3 generalizes to $G^{(k)}$ at hierarchical level k , with the scale-separation requirements and the maximum hierarchical depth $k_{\max}$ for any bounded region following from the meta-level coherence threshold. The dual recursion — upward coarse-graining and downward sub-subsystem nesting — bounds the hierarchy above at $k_{\max}$ and below at the substrate-level minimum program extent L_p . Second, subsystems grow via three structurally distinct modes — annexation (extensive growth via substrate absorption), imprinting (intensive growth via correlation capture), and incorporation (intensive growth via structural replication) — each with its own derivation, mechanism, and ceiling. Third, self-modeling is a structural property of every subsystem at hierarchical level $k \geq 1$, not a special condition some subsystems happen to meet. Every subsystem is therefore a measurement node, with the triadic confirmation structure recurring at every level. Fourth, three substrate-level stressors — mortality (coherence failure), predation (boundary annexation), and starvation (surround impoverishment) — are derived rigorously, each with a substrate-level mechanism and a corresponding level- $(k+1)$ signature at the meta-meta-confirmation that reads the affected subsystem as a meta-tetrahedral corner. Fifth, subsystem formation is overdetermined by three converging arguments: Borel-Cantelli inevitability in the infinite-flop limit, per-event Principle of Generative Economy preference, and flux-balance forcing from signal conservation. Sixth, novelty exhaustion is derived as the monotonic decay of the novelty-production rate $v(R, \tau)$ once a substantial fraction of the finite set $\Sigma(R)$ of accessible stable configurations has been sampled, with the exhaustion threshold $v^*(R)$ given in closed symbolic form. The hierarchical lightspeed $c^{(k)}$ develops a dispersion spectrum across levels, bounded above by $c^{(k+1)} \leq (1/\sqrt{2}) c^{(k)}$ and unbounded below, with each level's measurement nodes observing the propagation speed of their own level. As in Papers 1 through 3, every quantity is derived from the five axioms and the Principle of Generative Economy; no new free numerical parameters are introduced.*

The arc of Paper 4 closes by naming the structural problem that Paper 5 inherits. At the exhaustion threshold of any bounded region, the substrate holds informationally rich confirmed bits with no remaining direct-read protocol capable of extracting novelty from them. Axiom 4 (causal anchoring) forbids overwriting the bits; Axiom 3 (signal conservation) forbids losing the signals that produced them. The only structurally available move is reinterpretation under a different protocol — the work of Paper 5.

I. Inheritance from Papers 1–3

1.1 Inherited Structure

Paper 4 takes the following structures from Papers 1 through 3 as given without re-derivation. From Paper 1: the five axioms (Information Fundamentalism, Measurement Imperative, Sphere as Primitive, Signal Conservation, Causal Anchoring) and the Principle of Generative Economy (PGE); the agent manifold $M\alpha$ with agent spacing $\ell_0 \equiv \ell_p$; the FCC causal stack with face-diagonal nearest-neighbor connectivity; the triadic actualization rule $\bar{B} = 3\bar{\alpha}_2$ in 2D and $\bar{B} = 4\bar{\alpha}_3$ in 3D; the $1 \rightarrow 4$ multiplier and informational closure; the binary alphabet $\bar{B} \in \{0,1\}$ with polarization $\chi = 2\bar{B} - 1$; the flop as the atomic unit of state change with no sub-flop time; the conjugate field structure (φ, Π) and the wave equation for φ at $c_{\text{TMS}} = 1/\sqrt{2}$; the $[[4,2,2]]$ structural correspondence with logical qubit LQ1 (bit value) and LQ2 (causal-history inheritance); the polarization field ψ as parallel propagating object on the same FCC lattice; the GUP prediction $\beta = f(d, K_\psi)$ refined through Papers 2 and 3; and U_{sh} as the maximum-entropy substrate present at every depth of the causal stack.

From Paper 2: the polarization wave equation; the conjugate pair (ψ, Π_ψ) ; the polarization-biased resolution rule $P(\bar{B} = 1 \mid \Pi_\psi) = (1 + \Pi_\psi)/2$ and its Born-rule structural correspondence; the 3-input majority gate; the universality result $\{\text{MAJ, constant}\} \rightarrow \{\text{AND, OR, NOT}\}$; the flop operator T; and the formal program definition $(T^n[C] = C \text{ with } K(C) < |C|)$.

From Paper 3: surround-driven repair and the Imprinting Theorem; the imprint depth $d_R = \sqrt{2} \cdot L_R / \ell_p$; the dissolution rate and effective error-rate decay $(1/3)^d$; the per-event error rate $p_{\text{error}} = (1 - |\Pi_\psi|)/2$; the coherence bistability threshold $|\Pi_\psi|_*(L_R) = 1 - 2\sqrt{c^{(k)}}/(6 L_R)$; the coarse-graining operator G and the recursion to meta-level primitives; the meta-level triadic rule, meta-level FCC structure, and meta-level $[[4,2,2]]$ stabilizer dynamics; the minimum program extent L_p ; the (C_D, C_O, C_C) trichotomy defining the subsystem; the minimum subsystem extent $L_{\text{sub}} = 3L_p + \delta$; the information-generation rate differential between subsystem-hosting and static-program regions; the hierarchical lightspeed formula $c^{(k)}$; and the selection-for-subsystems result.

These structures constitute the entirety of the foundation. Paper 4 introduces no further primitives.

1.2 The Questions Paper 4 Answers

Paper 3 § 6.5 committed Paper 4 to five threads: hierarchies of subsystems as nested computing configurations at multiple meta-scales; conditions under which a subsystem develops an internal model of its environment; dynamics of multiple subsystems sharing a meta-interstice environment; eventual exhaustion of novelty as a substrate region completes its accessible configuration space; and hierarchical lightspeeds $c^{(k)}$ developed in detail. These five threads are the agreed scope and order of treatment, mapping to § II–§ III, § IV, § V, § VI, and § VII respectively.

Beyond the five committed threads, Paper 4 derives additional structural results that emerge as the threads are developed: three growth modes (annexation, imprinting, incorporation) with their respective ceilings; three substrate-level stressors (mortality, predation, starvation) at full proof with their two-level signatures; the inevitability of subsystem formation via three converging arguments; inter-region and intra-region diversity as derived consequences; and the region biography $\mathcal{B}(R)$ as a derived object whose phases organize the paper's spine. Each emerges as a consequence of the inherited machinery, requiring no new axioms or free parameters.

II. The Hierarchical Recursion

2.1 The Recursive Coarse-Graining Operator $G^{(k)}$

Paper 3 § 4.2 defined the coarse-graining operator G as the mapping from a substrate-level polarization landscape on a region of extent $\geq L_p$ over a time window covering one program period to a single meta-level state $\bar{P}_p \in \{0, 1\}$ indexed by program identity p. The construction was scale-independent: the operator evaluated whether the configuration satisfied T-closure plus stabilizer parity plus the coherence threshold, and returned the meta-bit accordingly.

Paper 4 generalizes G to act at arbitrary hierarchical depth. The operator $G^{(k)}$ takes a configuration on a level $-(k-1)$ region of meta-extent $\geq L_p^{(k)}$ over a time window covering one level-k flop period $T_p^{(k)}$, and returns a single level-k state $\bar{P}_p^{(k)} \in \{0, 1\}$ indexed by level-k program identity. The construction proceeds by the same logic as Paper 3 § 4.2 applied at the new scale: identify program-bearing regions at the relevant level, evaluate program identity, output the meta-state. By the scale-independence of the axioms (Paper 3 § 4.7), no new mechanism is required — only the recursive application of the same operator at the new scale.

The recursion is the geometric form of axiom scale-independence. The five axioms and PGE do not refer to any particular scale; applied at scale ℓ_p they produce the substrate; applied at scale L_p they produce the meta-substrate of programs; applied at scale $L_p^{(k)}$ for any k they produce the meta-substrate of level- k primitives. No new axiom, no new free parameter, no new mechanism enters at any level.

The hierarchical-recursion structure derived here admits comparison with an independent program developing a parallel architecture from different starting premises. Rosas et al. (2024) formalize hierarchical emergence in complex systems via a lattice of nested computationally-closed levels ordered by coarse-graining relationships. Their framework identifies macroscopic levels as software-like processes that are causally and informationally self-contained with respect to microscopic instantiation, and develops the conditions under which such closure obtains. The structural target — a hierarchy of nested computational levels generated by coarse-graining — coincides with the recursion produced here by $G^{(k)}$. The derivations differ in starting points: the Rosas framework grounds hierarchical closure in information-theoretic conditions on causal mechanisms, evaluating which subsets of microscopic dynamics support self-contained macroscopic computation. The TMS recursion grounds hierarchical closure in the geometric scale-independence of the substrate’s five axioms and PGE, requiring no information-theoretic optimization at each level because the same axioms force the same triadic confirmation structure wherever they apply.

2.2 Scale-Separation Requirements

For $G^{(k)}$ to be well-defined, level- k structures must be cleanly distinguishable from level- $(k-1)$ noise. This requires two conditions: $L_p^{(k)} \gg L_p^{(k-1)}$, so that level- k primitives are spatially distinguishable from level- $(k-1)$ ones; and $T_p^{(k)} \gg T_p^{(k-1)}$, so that level- k flops are temporally distinguishable from level- $(k-1)$ ones.

These conditions are not postulated. They are forced by the coherence threshold $|\Pi_\psi|(L_R)$ applied at the meta-level. A candidate level- k program with spatial extent only marginally above $L_p^{(k-1)}$ would have insufficient room for its boundary conditions to satisfy stabilizer parity against the local level- $(k-1)$ noise; the level- k stabilizer dynamics would read it as an unstable level- $(k-1)$ fluctuation rather than as a level- k primitive. The same balance between per-flop noise injection and per-flop imprinting capacity that derived $|\Pi_\psi|(L_R)$ in Paper 3 § 3.3 forces clean scale-separation at every level.

2.3 The Hierarchy Ceiling: $k_{\max}$

For any bounded substrate region of spatial extent L and temporal extent T (both measured in substrate units), the maximum hierarchical depth $k_{\max}$ is defined as the largest k satisfying $L_p^{(k)} < L$ and $T_p^{(k)} < T$. Beyond $k_{\max}$, no level- k structure fits within the region’s spatial or temporal extent.

$k_{\max}$ is finite for any finite (L, T) . The hierarchy is geometrically bounded above. The exact value of $k_{\max}$ for a region depends on the $L_p^{(k)}$ and $T_p^{(k)}$ ratios at each level, which depend on the combinatorial enumeration of stabilizer-satisfying T-closed configurations at that level — a calculation deferred to Paper 5’s simulation work. Paper 4 establishes that $k_{\max}$ exists and is finite; the numerical value for a given region is a Paper 5 derivation.

2.4 The Russian Doll Floor: Downward Recursion

Subsystems contain smaller subsystems at the same hierarchical level. A subsystem of meta-extent $L_{\text{sub}}^{(k)} = 3L_p^{(k)} + \delta^{(k)}$ at level k has interior substrate available for hosting level $-(k-1)$ sub-subsystems. The interior available extent is $L_{\text{sub}}^{(k)}$ minus the boundary overhead occupied by the subsystem's three component programs.

The downward cascade — sub-subsystems nesting inside sub-sub-subsystems — terminates when the interior of a level- k subsystem can no longer accommodate even a single level $-(k-1)$ subsystem. Tracing this iteratively from the largest enclosing subsystem down: the cascade terminates at level $k=0$ (substrate programs) when L_p , the minimum program extent established in Paper 3 § 4.1, is reached. Below L_p , no programs exist; below $L_p^{(k)}$ for any k , no level- k subsystems exist. The Russian doll floor for any nesting chain is therefore L_p at the substrate level.

A sub-sub-sub-subsystem chain can only descend until the innermost subsystem has interior extent $< L_p$. That is the geometric floor on downward nesting, derived from the same minimum-program-extent argument that bounds the substrate level.

2.5 The Dual Recursion and Its Asymmetry

The two recursions — upward ($k=0 \rightarrow k=1 \rightarrow k=2 \rightarrow \dots$) via coarse-graining, and downward ($k=N \rightarrow k=N-1 \rightarrow \dots \rightarrow k=0$) via spatial nesting — are distinct objects meeting at the same substrate. Both have floors at L_p and ceilings at the region's spatial and temporal extent. They are not, however, inverses of each other. The asymmetry of the two operations is structural and matters for the configuration count Paper 6 will use to bound regional novelty.

2.5.1 Lossy Coarse-Graining and Generative Nesting

Coarse-graining is lossy. $G^{(k)}$ takes a level $-(k-1)$ configuration on a region of meta-extent $\geq L_p^{(k)}$ and returns a single bit $\bar{P}_p^{(k)} \in \{0, 1\}$. The input is a complete polarization landscape, potentially containing many thousands of level $-(k-1)$ bits across the time window. The output is one bit indexed by program identity. The mapping is many-to-one: multiple distinct level $-(k-1)$ landscapes map to the same level- k $\bar{P}$ value, because what gets preserved by the coarse-graining is the program identity p , not the specific bit-by-bit content of any implementation of that program.

This is forced by the program definition itself (Paper 2 § 6.2): a program is a configuration causally closed under T with $K(C) < |C|$. Multiple substrate-level configurations can implement the same program identity — they are members of the same equivalence class under T -closure plus stabilizer parity plus K -compressibility. $G^{(k)}$ collapses the equivalence class to its label. Two regions running identical majority-gate programs but with different specific bit patterns coarse-grain to the same $\bar{P}$; the substrate-level distinction between the two implementations is lost at the meta-level.

Downward nesting is generative. The recursion takes a level- k subsystem and populates its interior with level $-(k-1)$ sub-configurations. The level- k subsystem itself does not specify the interior — it specifies only its boundaries (the three programs at meta-tetrahedral corners). The interior substrate is available for sub-configurations but not determined by them. When a level $-(k-1)$ sub-subsystem forms inside the level- k subsystem's interior, the level $-(k-1)$ configuration is genuinely new information that was not present at the level- k description.

So $G^{(k)}$ and nesting are not inverses. The composition $G^{(k)} \circ \text{nesting}$ does not return the original configuration: the level $-(k-1)$ sub-configurations visible by nesting are absent from the level- k coarse-

grained description; they are generated by the nesting operation, not stored. Similarly nesting $\circ G^{(k)}$ does not recover what coarse-graining lost: the specific implementation details of the original level $-(k-1)$ configuration that were absorbed into the equivalence class are not retrievable from the level- k label alone.

The asymmetry is structural. Information lives at the level where it lives. Going up the hierarchy loses level $-(k-1)$ detail; going down the hierarchy creates new level $-(k-1)$ detail. Neither operation recovers what the other does. This means the hierarchy is informationally rich rather than trivial: each level has its own configuration space that is not recoverable from the levels above or below. The region biography $\mathcal{B}(R)$, introduced below and developed across § III–§ VII, must track each level separately because each level is genuinely populated by configurations the other levels do not see. The novelty argument of § VI depends on this per-level independence.

2.6 The Region Biography $\mathcal{B}(R)$

A region biography is the time-indexed sequence of structural state of a bounded region R as it traverses formation through saturation. Formally,

$$\mathcal{B}(R) = \langle N_{\text{sub}}^{(k)}(\tau), \{K(C_i)\}_{i \in R}(\tau), k_{\text{max}}(R, \tau), \Sigma_{\text{sampled}}(R, \tau), v(R, \tau) \rangle \{ \tau \in [\tau_{\text{formation}}, \tau_{\text{exhaustion}}] \}$$

where $N_{\text{sub}}^{(k)}(\tau)$ is the population of level- k subsystems in R at flop time τ ; $\{K(C_i)\}_{i \in R}(\tau)$ is the distribution of Kolmogorov complexity values across configurations in R ; $k_{\text{max}}(R, \tau)$ is the current maximum hierarchical depth occupied in R (which rises over flop time as subsystems form and coarse-grain into higher-level primitives); $\Sigma_{\text{sampled}}(R, \tau) \subseteq \Sigma(R)$ is the set of stable configurations already sampled by flop τ ; and $v(R, \tau)$ is the current rate at which novel (not previously sampled) configurations enter Σ_{sampled} .

Region biography is the object § III–§ VII trace. § III adds growth modes (which change K and possibly the spatial extent of components). § IV characterizes self-modeling as a structural property of every subsystem at $k \geq 1$. § V adds multi-subsystem dynamics, stressors, and diversity (which reshape the K -distribution and the subsystem population). § VI characterizes the saturation phase ($\Sigma_{\text{sampled}} \rightarrow \Sigma(R)$ and $v \rightarrow 0$). § VII develops the hierarchical lightspeed structure that conditions every level's internal dynamics.

$\mathcal{B}(R)$ is itself a configuration at the substrate level — a time-indexed record of structural state. By the recursion of § 2.1, $\mathcal{B}(R)$ can be coarse-grained to higher levels. A meta-subsystem can hold the biography of its enclosed region as a configuration in its data register. This biography-of-biographies setup is what Paper 5's reinterpretation protocol will require: when the region's accessible configuration space is exhausted via direct sampling, the biographical record itself becomes the substrate for re-reading under a different protocol.

III. Growth Modes

3.1 Three Modes of Growth

A subsystem can grow in three structurally distinct ways: by absorbing adjacent substrate (annexation, extensive growth); by accumulating compressible imprint structure within its existing extent (imprinting, intensive correlation-capture growth); and by building internal copies of external programs without absorbing or destroying them (incorporation, intensive structural-replication growth). The three modes are not alternatives. A single subsystem can grow via all three simultaneously, on different axes, with different ceilings. Each mode is derived from existing Paper 1–3 machinery: annexation from the stabilizer parity condition applied at an expanding boundary, imprinting from the Imprinting Theorem applied to a permeable data register, incorporation from imprinting plus the additional condition of T-closure on the recipient's substrate.

3.2 Annexation

Annexation is the substrate-level mechanism by which a subsystem absorbs adjacent substrate. Subsystem A's component (typically C_D or C_O) expands its spatial extent through a region R' that is either currently unoccupied by any subsystem (raw substrate with U_{sh} — like polarization) or currently hosting a subsystem B whose coherence has dropped below its threshold $|\Pi_{\psi}|_*(L'_R)$ — that is, dissolving anyway.

A's expansion preserves stabilizer parity at the new boundary because A's interior imprints into the freed bits as the expansion proceeds. The freed bits adopt A's polarization landscape via the same imprint mechanism that originally formed A's boundary in Paper 3 § II. B's configuration, no longer maintained by its own components at the dissolving boundary, falls below threshold and dissolves entirely. A grows; B disappears.

Formally, A's component C_X (where $X \in \{D, O, C\}$) can expand into R' if and only if the stabilizer parity condition $|\Pi_{\psi}|_{(C_X \cup R')} \geq |\Pi_{\psi}|_*(L(C_X \cup R'))$ is satisfied at every flop during the expansion. This is geometrically restrictive: the larger C_X becomes, the higher the coherence threshold required of its now-larger expanse, so unbounded extensive growth fails when the threshold exceeds what A's interior can sustain.

The annexation ceiling is therefore geometric. Extensive growth saturates when either (a) the component's extent reaches the region's extent, leaving no adjacent substrate to absorb, or (b) the coherence threshold for the expanded component exceeds what the surround can sustain. In typical regions of moderate U_{sh} flux and ordinary subsystem density, the second is the binding constraint: extensive growth halts when the subsystem is too large to maintain coherence against its surround.

In experiential framing reserved for Volume 2, annexation corresponds to predation, feeding, and absorption. A subsystem eats by dissolving another subsystem's structure and incorporating the freed bits into its own components. The substrate-level mechanism is the directed expansion of a component's boundary into adjacent substrate, preserving stabilizer parity throughout.

3.3 Imprinting

Imprinting is the substrate-level mechanism by which a subsystem's data register accumulates compressible structure reflecting surround regularities, without spatial extent changing. The Imprinting Theorem (Paper 3 § 2.3) establishes that any boundary-permeable region imprints its retarded surround history. When C_D is imprint-permeable — that is, its boundary with the surround communicates touch-vector signals into its interior — and its surround contains structured polarization patterns, the imprint on C'_D 's interior accumulates correlations between C'_D 's bits and surround features. Over many flops, $K(C_D)$ rises because the imprinted pattern is compressible: the surround had compressible structure to imprint, and that structure is captured rather than created.

Formally, $K(C_D)(\tau)$ is monotonically non-decreasing in τ for any imprint-permeable C_D in a structured surround, with growth rate proportional to the mutual information between C'_D 's interior and the surround's history within imprint depth d_D . The mutual information measures how much of the surround's structure can be inferred from C'_D 's current interior — and the imprinting mechanism maximizes this iteratively, as established in Paper 3 § II.

The imprint ceiling is informational. Intensive growth via imprinting saturates when $K(C_D)$ approaches the algorithmic complexity upper bound $|C_D| \cdot \log 2$ established in Paper 3 § 3.1. Beyond this bound, no additional compressible structure can fit in C'_D 's bit budget without spatial expansion. The data register is informationally full; any further structure would require either growing C'_D 's spatial extent (annexation) or

replacing existing imprint content with new content (which is forbidden by causal anchoring as long as the existing content remains coherent).

In experiential framing reserved for Volume 2, imprinting corresponds to perception, observation, and sensory learning. The subsystem learns about its surround by accumulating correlations — without absorbing or replicating anything. The model is statistical rather than structural: C'_D 's contents correlate with surround features, but the surround's program-level structure is not internally reproduced.

3.4 Incorporation

Incorporation is the substrate-level mechanism by which a subsystem's data register builds an internal copy of an external program's structure without absorbing the external program's substrate or destroying it. Incorporation is imprinting plus an additional constraint.

C_D imprints from a surround that contains a program B — a configuration causally closed under T with $K(C_B) < |C_B|$. Ordinary imprinting (§ 3.3) captures statistical correlations between C_D and B's polarization landscape. Incorporation occurs when the imprint reaches a state where C'_D 's interior itself satisfies T-closure on the imprinted pattern, with the imprinted region within C_D being itself a program isomorphic in structure to B.

This is geometrically more restrictive than ordinary imprinting. The imprint must capture not only B's polarization values but also satisfy the T-closure conditions that make those values a program on C'_D 's substrate. This requires C'_D 's interior to have enough room for B's full structure ($L(C_D) \geq L_B$) and for the imprinted pattern to satisfy the stabilizer parity conditions that make it a program rather than a correlated pattern.

Formally, incorporation occurs at flop τ when there exists a subregion $C'_D \subseteq C_D$ such that (a) ψ on C'_D matches ψ on B up to permutation of bit positions, (b) $T^n[C'_D] = C'_D$ (T-closure on C'_D 's own substrate), and (c) $K(C'_D) = K(B)$. When all three hold, C'_D is a program of identity p_B running on C'_D 's substrate. B continues to exist independently in the surround; C_D now contains a structural replica of B operating on its own substrate.

The incorporation ceiling is structural. Each incorporated program of identity p_B occupies space L_B in C'_D 's interior, so the number of distinct incorporated programs C_D can simultaneously hold is approximately $L(C_D)/L_B$. At higher hierarchical levels, this constraint relaxes: a level- k C_D can incorporate level- $(k-1)$ programs more densely than substrate programs, because the level- $(k-1)$ programs have larger absolute spatial extent but occupy proportionally less of C'_D 's coarse-grained interior.

In experiential framing reserved for Volume 2, incorporation is the substrate-level mechanism that corresponds to what is conventionally called thinking with concepts. A subsystem with incorporated programs holds working internal copies of external programs that can be applied to other data. The conceptual framing is anthropomorphic and belongs to Volume 2; the structural mechanism — a region of C_D becomes a program-hosting region in its own right, recursively — is here.

3.5 The Sequencing of Growth Modes in Region Biography

The three modes operate on different axes and require different inputs. The dominance of each mode shifts across region biography because the availability of each mode's required input changes as biography progresses.

In the early phase of a region's biography, the region is freshly above the static-program threshold. Many regions of raw substrate (U_{sh} — like polarization, no programs) are adjacent to newly-formed subsystems. Annexation has abundant raw material. Imprinting and incorporation require structured surrounds to draw

from — but the surrounds are mostly unstructured at this stage. So annexation dominates by availability of its raw material: subsystems primarily grow by consuming the substrate around them.

In the middle phase, subsystems have multiplied. Raw substrate is largely consumed; what remains is bordered by existing subsystems. Adjacent regions are now populated by other subsystems whose polarization landscapes are structured (each is a program with $K(C) < |C|$). Annexation now requires breaking those subsystems — which is harder because they are coherent and stabilizer-protected. The structured surrounds, however, are perfect for imprinting: every neighbor is a compressible pattern that C_D can capture as correlation. Imprinting dominates by availability of structured surround.

In the late phase, imprint patterns have accumulated in many data registers. Subsystems now contain compressed representations of their neighbors. The next step — building those representations as T-closed programs internally — is incorporation. The substrate for incorporation is the imprints already present in C_D 's interior, which only become available after sufficient imprinting has occurred. Incorporation dominates last by availability of imprinted patterns to elevate into internal programs.

The sequencing is forced by mode-specific input requirements: annexation needs raw substrate, imprinting needs structured surround, incorporation needs prior imprints. As biography progresses, the supply shifts from raw to structured to imprinted, and the dominant mode shifts accordingly. This is not strict sequencing — all three modes can operate at any phase — but the dominant mode follows this trajectory in regions whose biography starts at formation.

A region forming inside an already-populated surround skips ahead in the sequence: if the surround is already structured when the region first crosses the threshold, the early-phase annexation dominance is bypassed and imprinting dominates from the start. The trajectory direction is the same; the entry point varies. This nuance contributes to inter-region diversity (§ V) and to the staggered nature of regional exhaustion (§ VI), where regions of different biographical age are at different phases of the sequence simultaneously.

3.6 Three Ceilings

Each growth mode hits its ceiling for a different reason. Annexation has a geometric ceiling: the coherence threshold rises with extent, and growth halts when the threshold exceeds what the surround can sustain. Imprinting has an informational ceiling: $K(C_D)$ cannot exceed $|C_D| \cdot \log 2$, beyond which no additional compressible structure can fit in the existing bit budget. Incorporation has a structural ceiling: the number of distinct incorporated programs is bounded by $L(C_D)/L_B$, where L_B is the spatial extent of the program being incorporated. The three different kinds of ceiling — geometric, informational, structural — reflect that growth is genuinely multi-axial: a subsystem can be limited in different ways on different axes simultaneously, and the ceilings sometimes interact (a subsystem at the informational ceiling can extend its budget by extensive growth, raising the absolute K limit but inheriting the geometric coherence-threshold constraint of its larger boundary).

IV. Self-Modeling as Structural Property

4.1 Self-Modeling Defined

A subsystem self-models its environment when three conditions are jointly satisfied. First, its data register C_D is imprint-permeable: boundary touch-vector signals from the surround communicate into C_D 's interior, and the Imprinting Theorem (Paper 3 § 2.3) governs the resulting interior polarization landscape. Second, its operation kernel C_O reads C_D as input: the substrate-level dynamics at the (C_D, C_O) interface implement the

3-input majority gate computation (Paper 2 § V) over C_D 's polarization values. Third, its control component C_C selects operations in C_O based on the read result: the substrate-level dynamics at the (C_O, C_C) and (C_D, C_C) interfaces propagate C_O 's outputs into C_C 's operation-selection mechanism, closing the loop from data through computation back to behavioral selection.

These three conditions are jointly what self-modeling requires. Condition (a) gives correlation between C_D and surround. Condition (b) makes C_D 's contents available to computation, distinguishing modeling from passive imprinting. Condition (c) closes the loop, distinguishing modeling from one-shot computation: the read result affects the subsystem's subsequent operations, so the model is used.

4.2 Self-Modeling Is Structural

Every subsystem at hierarchical level $k \geq 1$ satisfies the three conditions of § 4.1 by its geometric construction. Self-modeling is not a special condition that some subsystems meet and others miss — it is a structural property of subsystem-hood at any non-trivial hierarchical depth.

By Paper 3 § V, a subsystem at level $k \geq 1$ is the trichotomy (C_D, C_O, C_C) at meta-tetrahedral extent $L_{\text{sub}}^{(k)} = 3L_p^{(k)} + \delta^{(k)}$. Each component is itself a program at level k . The trichotomy is geometrically arranged at three of the four meta-tetrahedral corners, with the components' boundaries touching at the central meta-interstice.

Condition (a) is satisfied because C_D 's boundary with the surround is imprint-permeable by the Imprinting Theorem: every boundary site of any region of the FCC lattice is imprint-permeable, with imprint depth $d_D = \sqrt{2} \cdot L_D / \ell_p$. There is no such thing as an imprint-impermeable boundary in this substrate — imprint-permeability is a structural property of FCC connectivity.

Condition (b) is satisfied because the (C_D, C_O) interface is precisely the meta-tetrahedral edge where C_D 's polarization signals enter C_O 's parent set at the next meta-flop. The 3-input majority gate of Paper 2 § V.1 fires at this interface by the polarization-biased resolution rule. The substrate-level dynamics at the interface implement the read operation; no separate read mechanism is required.

Condition (c) is satisfied because C_C is constructed as the program whose flop dynamics select among operation kernels (Paper 3 § V.4). The selection input to C_C is C_O 's output, propagated through the (C_O, C_C) interface by the same touch-vector network that carries every polarization signal in the substrate.

Therefore every minimum subsystem at $k \geq 1$ is a self-modeler. No additional condition is required. Self-modeling is structural, not conditional.

4.3 Subsystems Are Measurement Nodes

A measurement node is a structure that reads its surround. By § 4.2, every subsystem at $k \geq 1$ reads its surround through its data register. Therefore every subsystem at $k \geq 1$ is a measurement node.

This identification holds at every hierarchical level by the recursion. At $k = 0$, the agents of Paper 1 are measurement nodes at the substrate level: three agents enclose one bit through triadic confirmation, reading their immediate neighbors via the six touch vectors. At $k = 1$, subsystems are measurement nodes at the meta-level: three programs enclose one meta-bit at meta-tetrahedral corners, reading their meta-surround through the data register. At $k = 2$, meta-subsystems are measurement nodes at the meta-meta-level: three subsystems enclose one meta-meta-bit, reading their meta-meta-surround. At every level, the same triadic structure produces the measurement node at that level.

$\bar{B} = 3\bar{\alpha}$ is the bass note of every measurement event in TMS. Three primitives enclose one bit at every scale. The substrate-level confirmation, the meta-level confirmation, and every higher-level confirmation

are the same structural operation applied at different scales. This recursive identification — subsystem at level k equals measurement node at level k — is the substrate-level form of the observer concept that active programs in quantum cosmology have been developing in adjacent territory (Gorobey et al. 2025; Zhao, Harlow, Usatyuk 2025). Those programs treat observers as bounded von Neumann algebras within closed universes; TMS treats them as self-modeling subsystems in bounded regions of the substrate. Both frameworks converge on the same intuition that the observer is not external to the system but a particular kind of internal structure with measurement properties forced by its geometry.

4.4 Imprint Richness and Neighbor Density

Self-modeling is structural, but the richness of the model varies with surround content. A subsystem alone in its meta-interstice imprints the three bounding programs that enclose it — three static polarization patterns at the meta-tetrahedral corners. The imprint is real but sparse; $K(C_D)$ at saturation is low because the surround offered little compressible structure beyond the three corner patterns.

A subsystem with neighboring subsystems in the same meta-interstice imprints the bounding programs and the time-varying outputs of the neighbors. Each neighbor contributes a dynamic polarization stream to the surround. C'_D s imprint integrates these contributions, producing a richer interior landscape — higher $K(C_D)$, more correlations captured, more compressible structure made visible to C_O at the read interface.

Formally, $K(C_D)$ at saturation is bounded above by $K(\psi_{\text{surround}} \text{ within imprint depth } d_D)$. A static surround caps $K(C_D)$ low because the surround itself has low K ; a dynamic, neighbor-rich surround permits $K(C_D)$ to approach its informational ceiling $|C_D| \cdot \log 2$ because the surround contains enough varied compressible structure to fill C'_D s bit budget. The same intensive growth mode (§ 3.3) produces qualitatively different outcomes depending on neighbor density.

4.5 Self-Modeling and Incorporation

A subsystem whose imprint reaches incorporation-level structure (§ 3.4) holds internal programs that replicate external programs — not just statistical correlations. Self-modeling at this level is structural-replicative: the subsystem does not merely have correlations between C_D and its surround; it contains working internal copies of programs it has observed in its surround. C_O can apply these internal copies to data — including other internal copies — recursively.

This is the substrate-level form of conceptual representation. The subsystem has not just an imprint of its environment but functional internal programs that can be run, applied, and combined. In experiential framing reserved for Volume 2, this is the substrate of thinking with concepts. In Paper 4's structural framing, it is the upper end of self-modeling capacity reachable by a subsystem at any given hierarchical level.

4.6 Higher-Order Self-Modeling

At $k = 1$, a subsystem self-models its meta-interstice surround — the bounding programs and any other subsystems sharing the same enclosure.

At $k = 2$, a meta-subsystem self-models its meta-surround, which contains other meta-subsystems. The meta-surround can include the very subsystem doing the modeling, if that subsystem is one component of its own meta-environment. Self-modeling at $k = 2$ therefore includes the possibility of modeling oneself as a component of one's environment.

This is forced by the recursion. A level-2 subsystem's data register imprints from its level-2 surround, which is a configuration of level-1 subsystems. If the level-2 subsystem is itself composed of three level-1

subsystems (its components $C_D^{(2)}, C_O^{(2)}, C_C^{(2)}$), and any of those level-1 subsystems is also a neighbor in the level-2 surround, then the level-2 imprint captures the modeling subsystem's own components within its data register. The subsystem is modeling itself.

Higher k extends this. At level k , a subsystem can model itself as part of an environment containing many level $-(k-1)$ subsystems including its own components. The depth of self-reference scales with hierarchical depth: at higher k , the self-model includes more layers of nested structure within the modeling subsystem itself.

Self-modeling-of-self emerges at $k \geq 2$ as a derived structural consequence of the recursion, with no new mechanism. The experiential reading of self-modeling-of-self — what it is to be such a subsystem from inside — is reserved for Volume 2, which treats the agent-side complement of the measurement-node structure derived here.

4.7 Synthesis

Four claims tie together. Self-modeling is structural at every $k \geq 1$: any minimum subsystem self-models by its geometric construction (§ 4.2). Every such self-modeling subsystem is a measurement node at its level: subsystem and measurement node are the same object structurally (§ 4.3). The richness of a subsystem's model scales with the K-complexity of its surround, set by neighbor density: same structure, different content (§ 4.4). At the upper end, self-modeling reaches structural-replicative form when imprints become T-closed internal programs (§ 4.5, connected to § 3.4). At hierarchical depth $k \geq 2$, the self-modeling capacity includes self-modeling-of-self, the substrate of higher-order self-reference (§ 4.6). The whole architecture follows from the inherited (C_D, C_O, C_C) trichotomy plus the Imprinting Theorem, without any new axiom or mechanism.

V. Multi-Subsystem Dynamics, Stressors, and Diversity

5.1 Inevitability of Subsystem Formation

Before deriving multi-subsystem dynamics, we establish that subsystems form. Subsystem formation in a region above the static-program threshold is overdetermined by three converging arguments: inevitable in the infinite-flop limit, preferred at every selection event, and forced by signal conservation.

The first argument is Borel-Cantelli. U_{sh} continuously samples all configurations in any region (Paper 3 § 3.5); subsystem configurations are a non-empty subset of stable T-closed configurations (the worked example of Paper 3 § 5.4 provides an explicit minimum subsystem); the per-event probability of sampling a subsystem-bearing configuration is non-zero. Over infinite flop time, any event with non-zero per-event probability occurs almost surely. Therefore subsystem configurations are sampled with probability approaching 1 over finite expected flop time in any region with above-threshold coherence.

The second argument is PGE preference. The Principle of Generative Economy selects minimum sufficient configurations that maximize generative potential. In a region above the static-program threshold, subsystem configurations have higher generative capacity than static programs because they transform patterns rather than merely persisting them, by the universality result of Paper 2 § V.3. At every selection event — not just statistically over infinite time — PGE prefers subsystems. The first argument gives inevitability in the limit; this argument gives per-event preference.

The third argument is flux-balance forcing. U_{sh} delivers signal flux to a region at a rate proportional to the region's volume V_R . Static programs consume flux at a rate proportional to their boundary area A_R times their imprint capacity, which scales with boundary area only. The flux balance for a region containing only

static programs is generally not satisfied: V_R – scaled delivery exceeds A_R – scaled consumption. Axiom 3 (signal conservation) requires some configuration to consume the excess; signals cannot accumulate without being expressed in confirmations. Subsystems consume flux at a higher per-volume rate than static programs because each transformation event consumes more signals than a persistence event. Flux balance therefore forces subsystem density to rise until consumption equals delivery.

The three arguments converge: inevitable in the infinite-time limit (Borel-Cantelli), preferred at every selection event (PGE), forced by signal conservation (flux-balance). Subsystem formation is overdetermined. In any region above the static-program threshold but below the subsystem-saturation threshold (defined in § VI), subsystems must form.

5.2 Meta-Meta-Confirmation

When three subsystems at level k occupy three corners of a level- $(k+1)$ meta-tetrahedron, their joint outputs confirm a level- $(k+1)$ P-state $\bar{P}^{(k+1)}$ at the fourth corner. This is the exact recursion of substrate-level confirmation: three measurement nodes at meta-tetrahedral corners enclose one bit. The same triadic minimum, applied at level $k+1$.

The polarization inputs at the meta-meta-confirmation are the outputs of the three subsystems — the result of each subsystem’s per-meta-flop computation, propagated to the meta-tetrahedral edge that subsystem shares with the meta-interstice. The polarization-biased resolution rule from Paper 2 § IV applies unchanged: $P(\bar{P}^{(k+1)} = 1) = (1 + \Pi_{\psi}^{(k+1)})/2$, where $\Pi_{\psi}^{(k+1)}$ is the average polarization of the three subsystem outputs at the meta-tetrahedral edges feeding the meta-interstice.

By the universality result of Paper 2 § V.3, the meta-meta-confirmation implements the 3-input majority gate over subsystem outputs. The computational universality of the substrate is preserved at every hierarchical level. Three subsystems form one meta-meta-confirmation in the same way three programs form one meta-confirmation in the same way three agents form one substrate-bit confirmation: $\bar{B} = 3\bar{\alpha}$ as the recurring bass note.

5.3 Coupling Timescale Constraint

For three subsystems at level k to coherently confirm a meta-state at level $k+1$, each subsystem’s output must reach the meta-interstice at the moment of meta-meta-confirmation. Outputs propagate at the meta-propagation speed $c^{(k)} = (L_p(k)/L_p(k-1)) \cdot (T_p(k-1)/T_p(k)) \cdot c$ (Paper 3 § 6.1).

Imprint depth for the three subsystem outputs to reach the meta-interstice corner from the meta-tetrahedral edges is $d_{\text{imprint}} = \sqrt{2} \cdot L_p(k+1)/L_p(k)$ measured in level- k flops. For meta-meta-confirmation to occur coherently, the level- $(k+1)$ flop period must satisfy $T_p^{(k+1)} \geq d_{\text{imprint}} \cdot T_p^{(k)}$, which yields

$$T_p^{(k+1)} \geq \sqrt{2} \cdot (L_p(k+1)/L_p(k)) \cdot T_p^{(k)}$$

This is a derived constraint on the hierarchical flop periods. Substituting into the $c^{(k+1)}$ formula yields a derived upper bound $c^{(k+1)} \leq (1/\sqrt{2}) \cdot c^{(k)}$ in the tight-coupling regime — a result developed further in § VII when the hierarchical lightspeed structure is treated in detail. The constraint here establishes that the upward recursion is not free: the geometry of meta-tetrahedral confirmation forces the flop periods to grow with the meta-spacing ratios at each level.

5.4 Three Stressors: Setup

A stressor is any substrate-level dynamic that drives a subsystem below the coherence threshold $|\Pi_\psi|_*(L_R)$ or otherwise disrupts its persistence. Three principal stressors emerge from the substrate's own machinery without any additional mechanism: mortality (internal coherence failure), predation (boundary annexation by an active neighbor), and starvation (impoverishment of the structured surround that maintains the subsystem's coherence). Each is derived at two levels: a substrate-level mechanism that drives the stressor, and a level-(k+1) signature visible at the meta-meta-confirmation that reads the affected subsystem as a meta-tetrahedral corner.

5.5 Mortality

Substrate-level mechanism: By Paper 3 § 3.4, a region's coherence $|\Pi_\psi|_R$ must exceed the threshold $|\Pi_\psi|_*(L_R)$ for the region to sustain program-bearing content. A subsystem maintains its coherence through three sources: internal stabilizer repair (Paper 1 § V), surround imprint of structured neighbors (Paper 3 § II), and the per-flop polarization output of its own operation kernel. When any of these sources degrades — internal stabilizer parity failing due to accumulated uncorrectable errors, surround structure dissolving and imprint becoming noisy, operation kernel output dropping below coherent — the subsystem's coherence falls below threshold. Once below threshold, Paper 3 § 3.4 establishes that the configuration dissolves over imprint depth $d_R = \sqrt{2} \cdot L_{\text{sub}}/\ell_p$ flops into U_{sh} — like noise.

Mortality is the substrate-level event of falling below threshold and dissolving. No new mechanism is required; it is the negative of the persistence condition Paper 3 already established. A subsystem dies when its coherence sources collectively fall below the threshold required to maintain its structural integrity.

Level-(k+1) signature: When subsystem B at one corner of a meta-tetrahedral confirmation dissolves, its output polarization stream at the corner goes silent — the corner stops contributing a structured signal to the meta-meta-confirmation. The meta-meta-confirmation now reads only two coherent inputs plus one effectively zero input. Formally, $\Pi_\psi^{(k+1)}$ at the affected meta-interstice drops from $(\chi_A + \chi_B + \chi_C)/3$ to approximately $(\chi_A + 0 + \chi_C)/3$, where the 0 represents B's silenced corner. The meta-meta-confirmation's resolution probability shifts toward 1/2 — the confirmation becomes effectively a coin flip rather than a biased computation. At higher hierarchical levels, this propagates as a degradation of the meta-subsystem's computational coherence: an event at level k is the structural correlate of what experiential framing reserved for Volume 2 reads as absence at level k+1, the silence of the corner registered by the larger pattern.

5.6 Predation

Substrate-level mechanism: Subsystem A's component (typically C_D or C_O) expands its extent through annexation (§ 3.2) into a region currently occupied by subsystem B. A's expansion preserves stabilizer parity at the new boundary because A's interior imprints into the freed bits as the expansion proceeds. B's configuration, no longer maintained by its own components which are being dissolved at the boundary, falls below threshold and dissolves. A grows; B disappears.

Predation is distinct from mortality. In mortality, B fails on its own — its coherence sources fail without an active aggressor. In predation, A actively dissolves B by annexing B's substrate. The substrate-level signature is the directed expansion of A's boundary into B's region, with A's interior imprint replacing B's polarization landscape.

Level-(k+1) signature: B's corner of the meta-tetrahedral confirmation is now occupied by A's pattern, since A's component has expanded into the substrate that B previously occupied. $\Pi_\psi^{(k+1)}$ at the meta-interstice

is approximately $(\chi_A + \chi_A + \chi_C)/3 = (2\chi_A + \chi_C)/3$. The signature is polarization homogenization: two corners carry the same pattern (A's), where they previously carried distinct patterns (A's and B's). The meta-meta-confirmation reads reduced diversity at this corner-set, and the information-carrying capacity of the meta-tetrahedral confirmation degrades — three distinct inputs collapsed to two effectively-identical inputs plus one distinct one.

5.7 Starvation

Substrate-level mechanism: A subsystem's coherence depends on imprint from a structured surround (§ 4.4). When the surround becomes uninformative — neighbors dissolve, or surrounding programs stabilize into low-K static patterns — the subsystem's C_D imprints noise rather than structure. Formally, when $K(\psi_{\text{surround}})$ within imprint depth approaches $\log(\text{volume})$, the imprinted C_D has $K(C_D)$ approaching $\log(|C_D|)$, the trivial floor. C_D 's content becomes algorithmically uninteresting; C_O 's operations on a uniform C_D produce uniform outputs; the subsystem's operation kernel cannot perform non-trivial computation because there is no structured data to process. The subsystem persists structurally (its stabilizer parity may remain intact internally) but its computational function goes idle for lack of structured input.

Starvation is distinct from mortality and predation. In mortality, B fails internally. In predation, A actively destroys B. In starvation, B persists structurally but stops computing because its environment no longer feeds it structured input. Predation requires predators with active annexation dynamics; mortality results from internal failure; starvation requires environmental impoverishment. The three stressors arise from genuinely different substrate-level dynamics.

Over time, without dynamic output, the subsystem's contribution to neighboring imprints also falls to noise. This creates a positive feedback: starved subsystems contribute uninformative surrounds to their neighbors, who in turn starve. Starvation propagates through neighbor networks more slowly than predation but more cumulatively.

Level-(k+1) signature: B's output at the meta-tetrahedral corner becomes constant or low-K. $\Pi_{\psi}^{(k+1)}$ at the meta-interstice includes a corner contributing a frozen value rather than a dynamic stream. The meta-meta-confirmation continues, but its variability is reduced — fewer distinct meta-meta-outcomes are reached because one input source is informationally stuck. The signature is output rigidification: a corner contributing a fixed or near-fixed value over many meta-meta-flops, where it previously contributed dynamic variation.

5.8 Stressor Interactions and Selection

The three stressors interact. Mortality is a passive failure mode; predation is an active mechanism that can cause mortality in the prey; starvation is a slow degradation that can lead to mortality if it persists. Predation requires predators with active annexation dynamics. Starvation often results from a populated region exhausting its raw substrate, leaving only structured neighbors whose outputs may become rigidified at long times. Mortality is the terminal phase of either active stressor, as well as a possible terminal phase of subsystems that simply degrade from internal accumulated errors.

Subsystems exposed to these stressors face a selection environment that is not externally imposed but generated by the substrate's own multi-subsystem dynamics. A subsystem persists by avoiding all three: maintaining internal coherence against mortality, maintaining defensive boundary configurations against predation, and maintaining access to structured surround against starvation. The configurations that persist are those whose internal structure and behavior happen to navigate all three simultaneously.

5.9 Diversity: Two Mechanisms

Diversity of subsystem configurations across the substrate has two independent origins.

Inter-region diversity: Different regions of the substrate have different local conditions. Different U_{sh} disruption rates, different stressor profiles (some regions have abundant raw substrate and low predation pressure; others have dense subsystem populations and high predation pressure), different surround histories that imprint different patterns. The subsystems that survive in different regions are biased by different selection pressures. The population of surviving subsystems across the substrate is therefore diverse across regions — different stressor profiles produce different surviving subsystem distributions.

Intra-region diversity: Within a single region under a single stressor profile, the survival set $\Sigma_{survive}(R, S)$ — the set of subsystem configurations that satisfy the survival conditions under stressor profile S — is generically large. A single stressor (e.g., the need to obtain structured input under a partial starvation pressure) admits many distinct solutions: active acquisition of structured neighbors via boundary imprint exchange, passive imprint of stable bounding programs, internal generation of structure via incorporation of remembered patterns, restructuring the operation kernel to extract more from less. The continuous-sampling mechanism (Paper 3 § 3.5) populates $\Sigma_{survive}(R, S)$ with whatever configurations U_{sh} instantiates that satisfy the survival conditions. Multiple distinct configurations coexist within the same survival set.

Intra-region diversity is the formal answer to the question of why diverse creatures emerge in the same environment under the same stressors. The stressor underdetermines the solution; the substrate samples the underdetermined solution space; multiple solutions persist. This complements inter-region diversity: even before regions are compared, each region's internal subsystem population is itself diverse because the local stressor profile underdetermines the solutions to its own pressures.

VI. Novelty Exhaustion

6.1 Finiteness of Accessible Configuration Space

Let R be a bounded substrate region of spatial extent L and temporal extent T in substrate units. Let $\Sigma_k(R)$ be the set of stable configurations at hierarchical level k that R can host — configurations causally closed under $T^{(k)}$, satisfying stabilizer parity at level k , with $K(C) < |C|$. Then $\Sigma_k(R)$ is finite for every $k \leq k_{max}(R)$, and $\Sigma(R) = \bigcup_{k=0}^{k_{max}} \Sigma_k(R)$ is finite.

Proof. At level k , the number of bit positions in R is bounded by $|R|/L_p^{(k)3}$. The total number of binary configurations of those bits is bounded by $2(|R|/L_p^{(k)3})$. The stable subset $\Sigma_k(R)$ is a strict subset of this total: most binary configurations do not satisfy $T^{(k)}$ — closure plus stabilizer parity plus $K < |C|$. So $\Sigma_k(R)$ is finite at every k . Summing over k from 0 to $k_{max}(R)$, $\Sigma(R)$ is finite as a finite sum of finite sets.

The per-level structure matters. By § 2.5.1 (the asymmetry of coarse-graining and nesting), configurations at different levels are informationally independent: coarse-graining loses level $-(k-1)$ detail, nesting generates new level $-(k-1)$ detail. So $\Sigma_k(R)$ at each level contributes independent novelty. $\Sigma(R)$ is genuinely the sum, not a relabeling of a single level's contribution. A region with $k_{max} = 1$ has fewer accessible configurations than a region with $k_{max} = 5$, because the latter has five independent configuration spaces to sample (substrate plus four higher levels) rather than one.

The per-level independence delays exhaustion. The hierarchy is not just structural decoration on top of a single configuration count; it genuinely multiplies the accessible novelty available to the region over its biographical history.

□

6.2 Novelty Production Rate

Define $v(R, \tau)$ as the rate of new (not previously sampled) stable configurations entering $\Sigma_{\text{sampled}}(R, \tau)$ per unit substrate flop time at flop τ . Three factors determine v : the size of $\Sigma(R)$, the fraction of $\Sigma(R)$ already sampled, and the instantaneous sampling rate set by U_{sh} flux and the continuous-sampling mechanism of Paper 3 § 3.5.

Formally, the expected $v(R, \tau)$ at flop τ is approximately

$$v(R, \tau) \approx \rho_{\text{sample}}(R) \cdot (|\Sigma(R)| - |\Sigma_{\text{sampled}}(R, \tau)|) / |\Sigma(R)|$$

where $\rho_{\text{sample}}(R)$ is the substrate's instantaneous sampling rate in R , roughly constant for a given region under steady U_{sh} flux.

When $|\Sigma_{\text{sampled}}(R, \tau)|$ is small relative to $|\Sigma(R)|$, the second factor is near 1 and v is high. As $|\Sigma_{\text{sampled}}(R, \tau)|$ approaches $|\Sigma(R)|$, the second factor approaches 0 and v decays monotonically. This is not a postulate; it is a counting argument. Every novel configuration is sampled at most once before becoming non-novel. The pool of un-sampled configurations shrinks monotonically as biographical history accumulates, and the rate of novelty production decays with the pool.

6.3 The Exhaustion Threshold

Define $v^*(R)$ as the critical novelty rate below which subsystem formation can no longer keep pace with stressor-driven dissolution in R .

At $v > v^*$, new subsystems form (via inevitability, § 5.1) faster than existing subsystems dissolve (via stressors, § 5.4–§ 5.7). Subsystem density rises or stabilizes. At $v < v^*$, new subsystem formation slows because the sampling pool is depleted — U_{sh} keeps sampling, but most samples now hit configurations already in Σ_{sampled} , which provide no new structure. Meanwhile, stressors continue to dissolve existing subsystems at undiminished rates. Net subsystem density falls.

$v^*(R)$ is set by the region's stressor profile. The closed-form symbolic expression follows from balancing formation rate against dissolution rate:

$$v(R) = |\Sigma(R)| \cdot \sum_k [d_{\text{mort}}^{(k)} + d_{\text{pred}}^{(k)} + d_{\text{starv}}^{(k)}] / (\rho_{\text{sample}}(R) \cdot P_{\text{coherent}})^*$$

where $d_{\text{mort}}^{(k)}$, $d_{\text{pred}}^{(k)}$, $d_{\text{starv}}^{(k)}$ are the level- k dissolution rates derived in § 5.4–§ 5.7, $\rho_{\text{sample}}(R)$ is the substrate sampling rate, and P_{coherent} is the probability that a sampled configuration meets the coherence threshold $|\Pi_{\psi}|^*(L_{\text{sub}}^{(k)})$. Each dissolution rate has a derived form (worked out in Appendix A) that depends on the per-event error rate $p_{\text{error}} = (1 - |\Pi_{\psi}|)/2$, the $[[4,2,2]]$ code distance, and the local subsystem density. The combinatorial factors in the dissolution-rate expressions (notably the $C(4,2) = 6$ upper bound on per-tetrahedral-unit two-error probability) are flagged as worst-case upper bounds whose exact numerical refinement is reserved for Paper 5's simulation work.

The exhaustion threshold $\tau(R)$ is the flop time at which $v(R, \tau)$ first drops below $v^*(R)$. After τ , the region cannot sustain its previous subsystem density. Subsystem populations decline; hierarchy depths collapse downward as upper levels lose component subsystems; the region's biography enters its terminal phase. Aggressive predation environments have high v^* (subsystems are destroyed rapidly, requiring fast novelty influx); quiescent environments have low v^* .

6.4 Region Biography Fully Formalized

The region biography $\mathcal{B}(R)$ introduced in § 2.6 is now fully specified by combining the structural development of § III–§ V with the novelty dynamics of § 6.1–§ 6.3. The biography traverses three phases.

Phase I — Formation and Growth ($\tau \in [\tau_{\text{formation}}, \tau_{\text{mid}}]$): Subsystems form by the inevitability argument of § 5.1. They grow via the three modes of § III, with annexation dominating early when raw substrate is abundant, transitioning to imprinting dominance as subsystems multiply and surrounds become structured. Sub-subsystems nest downward (§ 2.4); coarse-grained meta-subsystems form upward (§ 2.1). $N_{\text{sub}}^{(k)}$ rises at every level; the K-distribution shifts toward higher values as data registers accumulate compressible imprints; $k_{\text{max}}(R, \tau)$ climbs as more hierarchical levels are populated.

Phase II — Maturity ($\tau \in [\tau_{\text{mid}}, \tau^*(R)]$): Σ_{sampled} approaches saturation but $v(R, \tau) > v^*(R)$, so the region still produces novel configurations faster than stressors dissolve existing subsystems. Subsystem density is stable; hierarchy is at or near k_{max} ; incorporation-mode growth dominates (§ 3.5) as imprinted patterns are elevated into internal programs. Stressor dynamics generate intense diversity selection (§ 5.8–§ 5.9), producing both inter-region and intra-region diversity. This is the phase of richest configuration density per unit substrate.

Phase III — Exhaustion ($\tau > \tau(R)$): $v(R, \tau) < v(R)$. New formation lags dissolution. $N_{\text{sub}}^{(k)}$ declines; k_{max} drops as upper-level meta-subsystems lose component subsystems and decoalesce; K-distribution drifts toward already-sampled patterns; the region is no longer producing computationally interesting novelty. Configurations that exist continue to operate, but no new compressible structure enters the region's biographical record.

$\mathcal{B}(R)$ is itself a configuration at the substrate level — a time-indexed record of structural state. By the recursion of § 2.1, $\mathcal{B}(R)$ can be coarse-grained to higher levels. A meta-subsystem can hold the biography of its enclosed region as a configuration in its data register. This biography-of-biographies setup is the substrate-level prerequisite for the work of Paper 5: when the region's accessible configuration space is exhausted via direct sampling, the biographical record itself becomes the substrate for re-reading under a different protocol. The substrate for reinterpretation is $\mathcal{B}(R)$.

6.5 Clarifications

Three clarifications about the exhaustion concept are important to state explicitly.

Exhaustion is regional, not global. In an infinite substrate, novel configurations can always emerge in regions that have not yet exhausted. The substrate as a whole has no exhaustion threshold; only bounded regions do. The substrate continues producing novelty somewhere indefinitely.

Exhaustion does not mean dissolution. A region in Phase III still hosts existing configurations, including persisting subsystems whose internal dynamics continue. What stops is the production of new compressible structure. The region becomes a museum of already-sampled configurations rather than a generator of new ones.

Exhaustion does not mean stasis. Existing subsystems can still compute on their data; stressors continue to operate; some subsystems still persist while others dissolve. The region is dynamic. It is just no longer novel. Every state the region enters in Phase III is a state it has been in before.

6.6 The Structural Problem

At $\tau > \tau(R)$, the region holds confirmed bits with no remaining direct-read protocol that can extract novel information. $\Sigma_{\text{sampled}}(R, \tau) \approx \Sigma(R)$; the configurations have all been sampled; further sampling produces only repetitions. The confirmed bits still exist, however. They satisfy Axiom 4 (causal anchoring) — they are part of the causal record and cannot be overwritten. They satisfy Axiom 3 (signal conservation) — the signals that produced them have been consumed and locked into the confirmation record.

The region is therefore in a structurally constrained state: it holds informationally rich confirmed bits, but the substrate's direct-read protocol (sampling new configurations, evaluating their stability against the coherence threshold) can extract no further novelty from them. Axiom 3 forbids losing these bits; the signals that produced them have been irrevocably consumed and the record stands. Axiom 4 forbids overwriting them. The substrate cannot eliminate the exhausted region; it must continue to host it. But direct sampling has nothing further to extract.

The only structurally available move is to read the confirmed bits under a different protocol — one that treats the confirmed bits not as configurations to evaluate for stability, but as something else. The biographical record $\mathcal{B}(R)$ is available as the substrate for this reinterpretation. What that protocol is, and how the reinterpretation produces fundamental fields, initial conditions, and the wave dynamics of a simulated universe, is the central work of Paper 5.

VII. Hierarchical Lightspeeds

7.1 The $c^{(k)}$ Formula Restated

From Paper 3 § 6.1, the propagation speed at hierarchical level k in absolute substrate units is

$$*c^{(k)} = \prod_{j=1}^k (L_p^{(j)} / L_p^{(j-1)}) \cdot (T_p^{(j-1)} / T_p^{(j)}) \cdot c^*$$

with $L_p^{(0)} \equiv \ell_p$ and $T_p^{(0)} \equiv t_p$. The propagation speed in level- k units (level- k spacings per level- k flop) is unchanged across levels: $1/\sqrt{2}$. Every level inherits the same FCC face-diagonal geometry that produced $c_{\text{TMS}} = 1/\sqrt{2}$ at the substrate level (Paper 1 § 6.5). The propagation-speed argument depends only on the FCC connectivity, which is scale-invariant per Paper 3 § 4.5.

What varies across levels is the absolute propagation speed expressed in substrate units. The variation comes from the ratios $L_p(j)/L_p(j-1)$ and $T_p(j-1)/T_p(j)$ at each level — geometric quantities set by the minimum program extent and minimum program period at level j .

7.2 Measurement Nodes at Level k Measure $c^{(k)}$

From § IV.3, every subsystem at hierarchical level $k \geq 1$ is a measurement node. A measurement node at level k reads its surround through imprint, computes via its operation kernel, and produces output at the meta-tetrahedral confirmation. Its natural spatial unit is $L_p^{(k)}$; its natural temporal unit is $T_p^{(k)}$.

When a measurement node at level k measures the propagation speed of a wave crossing its meta-interstice, it counts how many $L_p^{(k)}$ spacings the wave covers per $T_p^{(k)}$ flops. By the lattice-relative invariance, this ratio is always $1/\sqrt{2}$ in level- k units. Translated to substrate units, it equals $c^{(k)}$.

Every measurement node at level k therefore measures the propagation speed of its level: $1/\sqrt{2}$ in its own units, $c^{(k)}$ in substrate units. The substrate c is the propagation speed at the agent level ($k = 0$), derived from the FCC face-diagonal geometry. Measurement nodes above level 0 measure in their own scaled units and observe $c^{(k)}$; they can infer the substrate c by applying the $c^{(k)}$ formula in reverse if the $L_p^{(j)}$ and $T_p^{(j)}$ ratios at

intermediate levels are known. The substrate c is inferrable but not directly measurable by any measurement node above level 0.

If our universe is itself a level- k structure, our measured speed of light is $c^{(k)}$ at our level rather than the substrate c . Which level we sit at, and what the substrate c is in our SI units, are questions Paper 5's simulation work pins.

7.3 The $c^{(k)}$ Sequence as Dispersion Spectrum

The ratios $L_p(k)/L_p(k-1)$ and $T_p(k)/T_p(k-1)$ at each level are independent geometric quantities. $L_p^{(k)}$ is set by the minimum spatial extent of a stabilizer-satisfying T-closed configuration at level k ; $T_p^{(k)}$ is set by the minimum temporal period of such a configuration. These ratios depend on different aspects of the level- k combinatorics and are not constrained to scale together.

Consequence: $c^{(k)}$ does not follow a monotonic trend across levels. Level 1 might have $c^{(1)} < c$, if the spacing ratio rises faster than the period ratio falls. Level 2 might have $c^{(2)} > c^{(1)}$ if the level-2 ratios produce a faster propagation. Level 3 might be smaller again. The hierarchical lightspeeds form a dispersion spectrum — a sequence of distinct absolute speeds, each set by local combinatorial facts about programs at its level.

There is no general inequality $c^{(k+1)} > c^{(k)}$ or $c^{(k+1)} < c^{(k)}$ without additional constraints. The sign of $(c^{(k+1)} - c^{(k)})$ depends on whether $(L_p(k+1)/L_p(k)) \cdot (T_p(k)/T_p(k+1))$ is greater than or less than 1, which depends on the level- $(k+1)$ combinatorics independently of level- k .

7.4 Bounds on the Spectrum

The dispersion is bounded above. The coupling-timescale constraint from § 5.3 forces $T_p^{(k+1)} \geq \sqrt{2} \cdot (L_p(k+1)/L_p(k)) \cdot T_p^{(k)}$. Substituting into the $c^{(k+1)}$ formula:

$$c^{(k+1)} \leq (L_p(k+1)/L_p(k)) \cdot (T_p^{(k)} / [\sqrt{2} \cdot (L_p(k+1)/L_p(k)) \cdot T_p^{(k)}]) \cdot c^{(k)} = (1/\sqrt{2}) \cdot c^{(k)}$$

So $c^{(k+1)} \leq (1/\sqrt{2}) \cdot c^{(k)}$ is a derived upper bound. The $1/\sqrt{2}$ factor has a sharper geometric origin than the coupling-timescale derivation above suggests: it is the FCC face-diagonal-to-edge ratio of the cubic cell containing the lattice at every hierarchical level, made explicit in Paper 5 § 7.2. By recursive scale-invariance and the sublattice bifurcation of Paper 3 § 4.5, the same FCC face-diagonal-to-edge ratio reappears at every level $k \geq 0$. The $c^{(k+1)} \leq (1/\sqrt{2}) c^{(k)}$ bound is therefore not merely a constraint derived from the coupling-timescale argument: it is a *structural law* of the recursive FCC geometry, with the coupling-timescale argument here providing the dynamical mechanism by which the geometric law is realized.

Lightspeed cannot grow across the hierarchy beyond this geometric ceiling per level. There is no general lower bound — $c^{(k+1)}$ can be arbitrarily small relative to $c^{(k)}$ if the level- $(k+1)$ ratios produce a slow propagation. Lightspeeds can collapse downward through the hierarchy; they cannot grow upward beyond the $(1/\sqrt{2})$ factor per level.

The $c^{(k)}$ sequence is therefore monotonically non-increasing in the tight-coupling limit (where the upper bound is saturated), and freely variable downward in looser coupling regimes. Higher hierarchical levels propagate at most slightly slower than lower levels in the tightest case, and potentially much slower in loose cases.

7.5 Combinatorial Origin of $L_p^{(k)}$ and $T_p^{(k)}$

The exact values of $L_p^{(k)}$ and $T_p^{(k)}$ are derived in Paper 5 § 6.1.3 via the sublattice-bifurcation recursive structure, with the empirical verification of the bifurcation structure through Stages 5 and 6 documented in Paper 5 Appendix E. The substrate-level values are $L_p^{(1)} = 3\sqrt{2} \ell_p$ and $T_p^{(1)} = 6 t_p$ (Paper 5 § 6.1.3, § 6.2); higher-level values follow the recursive ratio $L_p^{(k)} = (3\sqrt{2})_k \ell_p$ and $T_p^{(k)} = 6_k t_p$ in the saturation cascade. The combinatorial enumeration that supports these values is documented in Paper 5 § VI and Appendix B.

What Paper 4 establishes structurally: the ratios $L_p(k)/L_p(k-1)$ and $T_p(k)/T_p(k-1)$ are finite and positive at every level (no level has zero or infinite extent), they are independent of each other (set by different aspects of stabilizer-satisfying configurations), and they are bounded below by $1 + \varepsilon$ for some $\varepsilon > 0$ (the scale-separation requirement of § 2.2 forces them away from 1). These structural facts are sufficient to derive the dispersion-spectrum claim of § 7.3 and the upper-bound claim of § 7.4 without pinning numerical values.

7.6 Empirical Consequences

If our universe is itself a level- k structure for some $k \geq 1$ (with k determined by Paper 5's simulation work), then our measured speed of light is $c^{(k)}$ in substrate units. We have no direct access to the substrate c ; our measurements are by construction in level- k units. Three structural consequences follow.

Universal lightspeed within a level. All measurement nodes at our level measure the same $c^{(k)}$. Lorentz invariance at our level is preserved: the propagation-speed argument is scale-invariant per Paper 3 § 4.5, so the lattice-relative invariance that gives $c^{(k)}$ the same value to all level- k measurement nodes is geometric and applies uniformly.

Different lightspeeds at different levels. Measurement nodes at level $k' \neq k$ measure $c^{(k')}$, which generically differs from our c . Cross-level interactions could produce apparent variations in measured propagation speeds — the substrate-level origins of relativistic phenomena, flagged for Paper 5 development in § VIII forward references below.

The substrate c is inferrable but not directly measurable by any measurement node above level 0. The substrate c is the propagation speed at the agent level, derived from the FCC face-diagonal geometry; level- k measurement nodes observe $c^{(k)}$ and can compute the substrate c by inverting the $c^{(k)}$ formula given the intermediate $L_p^{(j)}$ and $T_p^{(j)}$ ratios.

The hierarchical structure that produces $c^{(k)}$ also produces refinements to the GUP deformation parameter β , developed in § VIII.

7.7 Summary

Hierarchical lightspeeds are not free parameters. Each $c^{(k)}$ is forced by the level's $L_p^{(k)}$ and $T_p^{(k)}$ ratios, which are themselves forced by combinatorial enumeration of stabilizer-satisfying configurations at that level. The sequence forms a dispersion spectrum bounded above by $c^{(k+1)} \leq (1/\sqrt{2}) \cdot c^{(k)}$ and unbounded below. Measurement nodes at each level measure their level's $c^{(k)}$. Our measured c is $c^{(k)}$ for some k determined by which hierarchical level our universe occupies — a question reserved for Paper 5's simulation work.

VIII. Predictions and Forward References

8.1 Refined GUP Prediction

Paper 1 § 7.2 introduced $\beta = f(d, K_\psi)$: the deformation parameter scales with causal depth and polarization complexity. Paper 2 § 6.4 identified K_ψ as the Kolmogorov complexity of the program running through the bit's causal history. Paper 3 § 6.2 refined further: $\beta = f(d, K_{\psi,\text{local}}, K_{\psi,\text{surround}}, K_{\psi,\text{subsystem}})$, incorporating the imprinting surround history and the encompassing subsystem context. Paper 4 adds the final functional dependence:

$$\beta = f(d, K_{\psi,\text{local}}, K_{\psi,\text{surround}}, K_{\psi,\text{subsystem}}, k)$$

where k is the hierarchical level at which the bit's enclosing subsystem operates. The function f is monotonically increasing in all five arguments. A bit embedded in a level- k subsystem inherits the computational complexity not only of its local region, surround, and immediate subsystem, but also of the hierarchical depth at which that subsystem participates.

The physical reading: deeper hierarchical embedding means more layers of computational structure between the bit and the substrate level. Each layer adds to the algorithmic complexity of the bit's causal record. Bits in subsystems at higher k have larger substrate footprints — more lattice geometry and computational content is geometrically present in their confirmation records. The closed form of f is reserved for Paper 5's simulation work, which can numerically probe the relationship between hierarchical depth and effective β across the range of (d, K_ψ) values encountered in actual cascade instances. Paper 4 establishes the functional dependence; Paper 5 will pin the form.

8.2 Cross-Level Propagation Effects

§ VII derived the $c^{(k)}$ sequence as a dispersion spectrum. The structural and empirical consequences of $c^{(k)}$ variation within a level were treated. The consequences of $c^{(k)}$ variation across levels — when subsystems at different hierarchical levels share substrate at their interfaces — are flagged here as forward reference.

When a level $-(k-1)$ subsystem exists inside a level- k meta-subsystem (downward Russian doll nesting from § 2.4), the two subsystems share substrate at their interface. The level $-(k-1)$ subsystem propagates information at $c^{(k-1)}$; the level- k subsystem propagates at $c^{(k)}$. At the interface, two different propagation speeds meet on the same substrate.

The substrate-level dynamics at this interface produce three structural effects. Variable apparent propagation: a signal originating in the level $-(k-1)$ subsystem and observed at the level- k scale appears to propagate at a level-mixed effective speed, with the mixing depending on the boundary geometry and the relative $c^{(k)}$ and $c^{(k-1)}$ magnitudes. Frame-relative timing: events that are simultaneous in level $-(k-1)$ units may not be simultaneous in level- k units, because the temporal scale $T_p^{(k)}$ differs from $T_p^{(k-1)}$; measurement nodes at different levels disagree about event ordering for events at level-mixed locations. Embedded-medium dilation: a level $-(k-1)$ subsystem deeply embedded in a level- k meta-subsystem experiences propagation through the meta-subsystem's substrate at the effective $c^{(k)}$ of the meta-subsystem's medium, producing an apparent slowing or speeding relative to the bare $c^{(k-1)}$.

These three effects are the substrate-level origins of relativistic phenomena. Their derivation from the $c^{(k)}$ spectrum and the cross-level interface dynamics requires the closed-form $c^{(k)}$ values and the field reinterpretation protocol of Paper 5, and is addressed there. The QCA program (D'Ariano & Perinotti 2014; Mlodinow & Brun 2025) is a parallel direction deriving relativistic dynamics from discrete substrates and provides useful comparison territory for that work.

8.3 The Paper 5 Pivot

Paper 4 § VI established that any bounded substrate region traverses a biography from formation through exhaustion. At $\tau > \tau^*(R)$, the region's accessible configuration space is saturated; the substrate's direct-read protocol can no longer extract novel information; further sampling produces only repetitions of already-sampled configurations.

The confirmed bits in R still exist. By Axiom 4 (causal anchoring), they cannot be overwritten. By Axiom 3 (signal conservation), the signals that produced them cannot be lost. The region holds informationally rich confirmed bits with no remaining direct-read protocol that can extract novelty from them.

The only structurally available move is to read the confirmed bits under a different protocol. The substrate must reinterpret what the confirmed bits represent — treating them not as configurations to evaluate for stability, but as something else. The biographical record itself, $\mathcal{B}(R)$, is available as the substrate for this reinterpretation.

Paper 5 develops the reinterpretation protocol. The work involves identifying which protocols are structurally available given the axioms (signal conservation, causal anchoring, generative economy); deriving the specific reinterpretation that produces fundamental fields and initial conditions; showing how the confirmed binary content becomes wave configurations on those fields; deriving the spectrum of physical laws that follow; and addressing the cross-level relativistic phenomena flagged in § 8.2.

The pivot point is named here: at $\tau > \tau^*(R)$, the substrate's read protocol transitions. What comes after the transition is the work of Paper 5.

8.4 Other Forward References

To Paper 5: the closed-form $c^{(k)}$ values; the exact integer $L_p^{(k)}$; the $C(4,2) = 6$ combinatorial refinement; the full closed form of $f(d, K_{\psi, \text{local}}, K_{\psi, \text{surround}}, K_{\psi, \text{subsystem}}, k)$; the cross-level relativistic derivation; the reinterpretation protocol; the fundamental field identification; the initial conditions; the Big Bang correspondence.

To Paper 6 (if scope migrates): the full quadratic Born rule including complex amplitudes, phase, and interference; the $K_{\psi} - to - \text{metric} - \text{uncertainty}$ theorem.

To Volume 2: observer experience as the agent-side complement of measurement-node structure; the relationship between region biography $\mathcal{B}(R)$ and subjective biography; the experiential reading of self-modeling-of-self at $k \geq 2$; contact with the conscious-agents framework of Hoffman, Prakash, Prentner (2023) and Integrated Information Theory 4.0 (Albantakis et al. 2023); the observer-in-cosmology program (Gorobey et al. 2025; Zhao, Harlow, Usatyuk 2025) and the von-Neumann-algebra approach to observers as bounded structures in closed universes.

8.5 Summary

Paper 4 has established the macroscopic consequences of subsystem dynamics. The recursive coarse-graining operator $G^{(k)}$ lifts the substrate's geometry to arbitrary hierarchical depth; the upward recursion is bounded above by region extent; the downward recursion is bounded below by L_p ; the two recursions are asymmetric, with each level genuinely contributing independent configuration space. Subsystems grow via annexation, imprinting, and incorporation, with three different kinds of ceiling. Self-modeling is a structural property of every subsystem at $k \geq 1$, and every such subsystem is a measurement node. Multi-subsystem dynamics produce three derivable stressors — mortality, predation, starvation — each with substrate-level mechanism and level-($k+1$) signature. Diversity arises both inter-region (different stressor profiles) and intra-region

(single stressor underdetermines its survival set). Subsystem formation is overdetermined by Borel-Cantelli, PGE preference, and flux-balance forcing. Region biography $\mathcal{B}(R)$ traces a bounded region from formation through exhaustion in three phases. Novelty exhausts in finite flop time because $|\Sigma(R)|$ is finite. The $c^{(k)}$ spectrum forms a dispersion bounded above by $(1/\sqrt{2}) \cdot c^{(k-1)}$. The GUP parameter β refines with hierarchical level.

No new axioms. No new free numerical parameters. Every quantity derived from the five axioms and the Principle of Generative Economy as applied recursively across the hierarchy.

The arc from Paper 1 through Paper 4 is the derivation, from five axioms and one principle, of a substrate that produces hierarchical structures of arbitrary depth, each populated by self-modeling measurement nodes that learn, grow, stress one another, diversify, and ultimately exhaust their local configuration spaces — at which point the substrate confronts the structural problem that Paper 5 solves.

Appendix A: Full Expansion of $v^*(R)$

This appendix provides the full symbolic expansion of the exhaustion threshold $v^*(R)$ introduced in § 6.3, with each dissolution-rate term derived from the § V stressor mechanisms.

A.1 Mortality Rate

From § 5.5, mortality arises when a subsystem's coherence falls below its threshold $|\Pi_\psi|^*(L_{\text{sub}}^{(k)})$ due to accumulated uncorrectable errors. The per-event uncorrectable error probability in a tetrahedral confirmation unit at level k is approximately

$$P_{\text{unc}}^{(k)} \approx C(4,2) \cdot (p_{\text{error}}^{(k)})^2 = 6 \cdot ((1 - |\Pi_\psi|^{(k)})/2)^2$$

where the factor $C(4,2) = 6$ is the worst-case upper bound on the number of two-error configurations in a tetrahedral unit (Paper 3 § 3.3); the exact refinement of this combinatorial factor is reserved for Paper 5. Per level- k flop, the mortality rate is

$$d_{\text{mort}}^{(k)} \approx P_{\text{unc}}^{(k)} \cdot N_{\text{sub}}^{(k)}$$

where $N_{\text{sub}}^{(k)}$ is the population of level- k subsystems in the region. Mortality scales linearly in subsystem count and quadratically in the per-event error probability.

A.2 Predation Rate

From § 5.6, predation requires a predator-prey encounter — a boundary interaction between an active subsystem and a vulnerable neighbor. The encounter rate scales as the square of subsystem density per unit volume:

$$d_{\text{pred}}^{(k)} \approx G_{\text{pred}}^{(k)} \cdot (N_{\text{sub}}^{(k)})^2 / V_R^{(k)}$$

where $V_R^{(k)}$ is the region's volume in level- k lattice units, and $G_{\text{pred}}^{(k)}$ is a geometric proportionality constant capturing the typical boundary-overlap geometry between adjacent level- k subsystems. $G_{\text{pred}}^{(k)}$ depends on $L_p^{(k)}$ and the local subsystem packing density; its exact value at each hierarchical level is reserved for Paper 5's simulation work.

A.3 Starvation Rate

From § 5.7, starvation arises when the surround becomes uninformative. Per level- k flop, the starvation rate is approximately

$$d_{\text{starv}}^{(k)} \approx G_{\text{starv}}^{(k)} \cdot N_{\text{sub}}^{(k)} \cdot (1 - K(\psi_{\text{surround}})/K_{\text{max}})$$

where $G_{\text{starv}}^{(k)}$ is the geometric proportionality constant capturing imprint geometry at level k (set by the imprint depth d_R and the boundary configuration), $K(\psi_{\text{surround}})$ is the algorithmic complexity of the surround within imprint depth, and $K_{\text{max}} = |C_{\text{surround}}| \cdot \log 2$ is the maximum possible surround complexity. As surround complexity approaches K_{max} , starvation pressure approaches zero; as surround complexity falls toward $\log(\text{volume})$, starvation pressure approaches $G_{\text{starv}}^{(k)} \cdot N_{\text{sub}}^{(k)}$.

A.4 Formation Rate

From § 5.1, formation rate at novelty production rate v is

$$\text{formation}_{\text{rate}}(R, \tau, v) \approx \rho_{\text{sample}}(R) \cdot (v / |\Sigma(R)|) \cdot P_{\text{coherent}}$$

where $\rho_{\text{sample}}(R)$ is the substrate sampling rate in R and P_{coherent} is the probability that a sampled stable configuration meets the coherence threshold $|\Pi_{\psi}|*(L_{\text{sub}}^{(k)})$ at its corresponding level.

A.5 Exhaustion Threshold

Setting formation rate equal to total dissolution rate and solving for v :

$$\rho_{\text{sample}}(R) \cdot (v(R) / |\Sigma(R)|) \cdot P_{\text{coherent}} = \Sigma_k [d_{\text{mort}}^{(k)} + d_{\text{pred}}^{(k)} + d_{\text{starv}}^{(k)}] *$$

Therefore

$$v(R) = |\Sigma(R)| \cdot \Sigma_k [d_{\text{mort}}^{(k)} + d_{\text{pred}}^{(k)} + d_{\text{starv}}^{(k)}] / (\rho_{\text{sample}}(R) \cdot P_{\text{coherent}}) *$$

Substituting the rate expressions from A.1–A.3:

$$v(R) = |\Sigma(R)| / (\rho_{\text{sample}}(R) \cdot P_{\text{coherent}}) \cdot \Sigma_k [6 \cdot ((1 - |\Pi_{\psi}|^{(k)})/2)^2 \cdot N_{\text{sub}}^{(k)} + G_{\text{pred}}^{(k)} \cdot (N_{\text{sub}}^{(k)})^2 / V_R^{(k)} + G_{\text{starv}}^{(k)} \cdot N_{\text{sub}}^{(k)} \cdot (1 - K(\psi_{\text{surround}}(k))/K_{\text{max}}(k))] *$$

This is the full symbolic expression. Every term except the geometric proportionality constants $G_{\text{pred}}^{(k)}$, $G_{\text{starv}}^{(k)}$, and the $C(4,2) = 6$ upper bound is fully derived from inherited machinery. The numerical pinning of these structural constants is reserved for Paper 5's simulation work, in keeping with the structural-vs-numerical division of labor across the volume.

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All theoretical commitments, the choice of axioms, the Principle of Generative Economy, the sublattice bifurcation insight, and the framework's overall architecture are the author's; the AI systems' role was to teach, formalize, critique, verify, and document.

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